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RESEARCH ARTICLE

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Sex determines the natural killer cell-mediated immunity against pancreatic cancer

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ABSTRACT

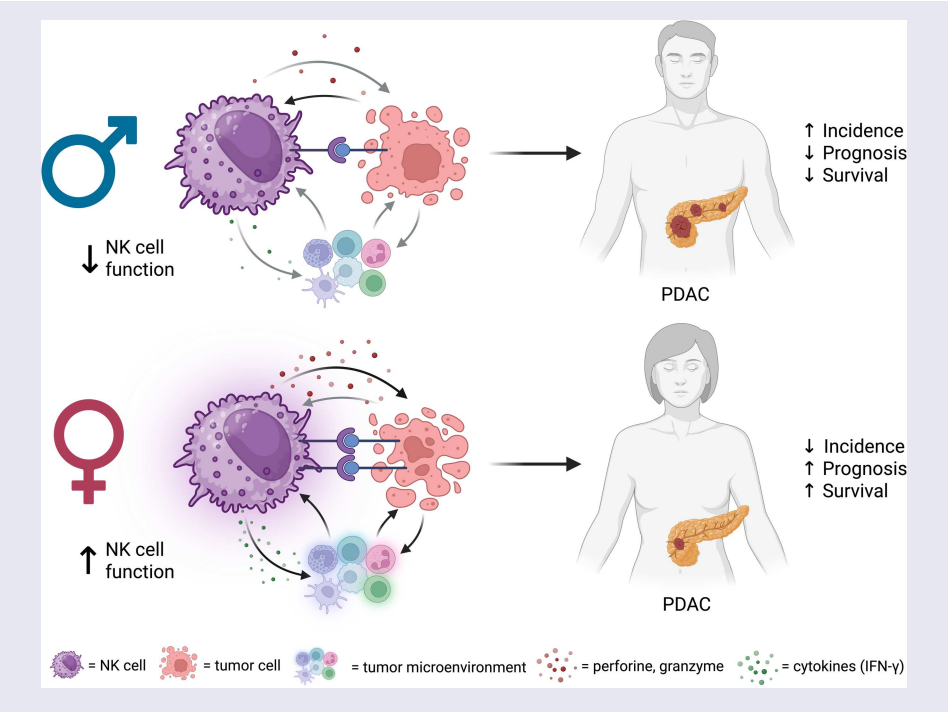
Pancreatic ductal adenocarcinoma (PDAC) ranks among the most aggressive malignancies and exhibits a higher prevalence and poorer prognosis in male patients. While both the importance of natural killer (NK) cells in anti-tumor immunity and sex differences in NK cell function are well established, the interplay between biological sex and NK cell-mediated responses in PDAC has not been systematically addressed. Here, we provide novel insights revealing that both the intrinsic cytotoxic capacity of NK cells and the tumor microenvironment differ between sexes and jointly modulate anti-tumor immunity in PDAC. Using *in vitro* assays with primary NK cells from male and female mice, we found that female-derived NK cells exhibit consistently superior cytotoxic activity and immune activation. Importantly, these findings were reflected and even amplified *in vivo*. In a syngeneic allograft PDAC model, female mice displayed enhanced NK cell function, and female tumors exhibited distinct structural characteristics supporting stronger NK cell activity. Notably, this occurred despite the use of the same cell line in both sexes, indicating that host sex significantly shapes the composition and architecture of the tumor microenvironment. Our results provide new evidence that sex differences in NK cell-mediated immunity, within the broader immune and stromal landscape, may differentially impact tumor control in male and female PDAC hosts. This study therefore underscores the importance of considering sex as a biological variable in the preclinical evaluation of pancreatic cancer therapies.

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Introduction

Natural killer (NK) cells are granular innate lymphocytes that represent an efficient endogenous first-line defense against cancer. They can identify virus-infected and malignant cells without prior antigen exposure, distinguishing them from cytotoxic T cells.[1] Upon degeneration of an individual cell, it is recognized by NK cells and subsequently eliminated prior to further proliferation. Nevertheless, NK cells can infiltrate the tumor microenvironment (TME) of solid cancers as well as tumor-infiltrated lymph nodes or metastases, therefore playing an important role in preventing tumor growth and spread.[2,3] When activated, NK cells can perform two different killing approaches: the release of lytic granules containing perforin and granzyme, or inducing death receptor-mediated apoptosis via the expression of Fas ligand or TRAIL.[4] In addition, NK cells are major producers of IFN-γ, as well as other cytokines like TNF-α and GM-CSF, that attract and activate other immune cells.[5] The NK cell activity is determined by the balance of inhibitory and activating receptors on the surface, which in turn are affected by activating and inhibitory signals.[6] NK cells can recognize whether a potential target cell exhibits a ligand profile associated with oncogenic transformation.[7] One of the strongest NK cell activators is the lack of MHC class I-specific receptors on a target cell.[8] Also, NK cells recognize stress-induced ligands.[9] In addition to surface molecules, soluble components of the TME impact the NK cell function.[10] The joint activating and inhibitory signals determine whether a target cell is killed.[7]

Women in general tend to experience better outcomes than men when affected by conditions such as infectious diseases, sepsis, trauma, or injury.[11,12] However, this is accompanied by a markedly higher prevalence of autoimmune diseases among women compared to men.[13] Sex differences in NK cell function have already been described previously. The general consensus is that female NK cells might be more functional than male NK cells.[14] Prior studies have proposed hormonal regulation as well as epigenetic differences as reasons for that. For instance, estrogens in general are known to exhibit immunoenhancing effects.[15] Previous research suggests that estrogens' impact is dose-dependent,[16] and their effect on NK cells may vary depending on the NK cell phenotype.[14] On the other hand, testosterone exhibits mainly immunosuppressive effects on NK cells.[14,17] However, the inhibitory impact is not fully unraveled, but might involve the upregulation of PD-L1.[14] In line with this, medical castration in healthy men has been proven to increase circulating numbers of NK cells.[18] Sex hormones exert their effects not only through classical receptors such as ESR (estrogen receptor)

and AR (androgen receptor) but also via alternative pathways, involving various GPRs or ZIP9.[19,20] Additionally, testosterone can be converted intracellularly to DHT (dihydrotestosterone), further interacting with distinct pathways, and even ESRs.[21,22]

Epigenetic mechanisms influencing NK cell activity involve an increasing array of genes and microRNAs, which are partially X-linked. Naturally existing variations within these genetic elements can explain sex-based disparities in immune system characteristics.[13,23] Additionally, as females have two X chromosomes, one of them is randomly silenced during X chromosome inactivation to avoid double dosage of proteins in females.[23] However, some genes can escape X-inactivation and are expressed from both chromosomes.[24] This results in elevated levels of functional immune proteins in females, potentially leading to a better immune function.[23] Recently, the X-linked histone demethylase UTX (*Kdm6a*) was identified to escape X inactivation and shape gene expression, leading to enhanced NK cell responsiveness and function in females.[25]

With over 400,000 deaths per year, pancreatic ductal adenocarcinoma (PDAC) is one of the deadliest tumor entities worldwide.[26] The incidence and mortality are increasing, and PDAC is predicted to be the second-leading cause of cancer-related death by 2030.[27] The pathophysiology and progression of PDAC are strongly affected by the TME.[28] NK cells in the peripheral blood and in the tissue of PDAC patients exhibit a reduced cytotoxic function and a regulatory, IL-10-producing phenotype.[29] Patients with high NK cell infiltration in the tumor demonstrated improved survival, compared with those with low NK cell infiltration.[30] TME-infiltrating NK cells display downregulated activating receptors caused by immune evasion factors produced by tumor cells, e.g., tumor-derived lactate and expression of IGHG1.[31,32] Furthermore, tumor cells consume important nutrients, such as vitamin B6, which NK cells need as a critical energy source to exhibit proper function.[30] As there are various approaches to increasing the NK cell antitumor activity by immunotherapy, more fundamental research promises major advances in the treatment of PDAC.[7]

The biological sex plays a crucial role in multiple tumor entities, like melanoma, lung cancer, and bladder cancer,[33-36] and is also known to highly influence PDAC pathology. Men have higher incidence and mortality rates, poorer prognoses, shorter survival times after diagnosis, and poorer therapy outcomes.[37-39] Since the early development of PDAC is closely related to the TME, it is conceivable that sex-specific differences in the NK cell immune response play a crucial role in the development and progression of pancreatic cancer.[28,40] In line with this, stronger innate immune responses have been described in female PDAC patients.[41]

The present study aims to answer the question of whether biological sex influences the NK cell-mediated immunity against PDAC and can, therefore, at least partially explain the divergent clinical outcomes. We systematically investigated sex-dependent differences in NK cell-mediated antitumor responses in PDAC. Using complementary *in vitro* and *in vivo* approaches, we assessed functional, transcriptional, and phenotypic differences between male and female NK cells and their impact on tumor control. In addition, we applied high-dimensional, multiplex immunofluorescence-based analyses in combination with RNA-sequencing of syngeneic PDAC allografts to characterize sex-specific differences in tumor growth, immune infiltration, cellular activity, epigenetic regulation, and stromal composition. Together, this integrative approach was designed to elucidate how biological sex shapes NK cell function toward PDAC cells *in vitro* and influences PDAC growth *in vivo* in concert with other components of the TME.

Materials and methods

Cells and culture conditions

Primary NK cells were isolated from the mouse spleen and utilized immediately for experiments after isolation owing to their limited *ex vivo* lifespan. For co-culture experiments, the culture medium of the target cell line was used.

Stable GFP-transfected mouse Panc02 cells (GenTarget Inc.) were cultivated in RPMI supplemented with 10% FBS and 1% penicillin–streptomycin. To select GFP-positive cells, 3 µg/ml puromycin was added to the cell culture flask.

The mouse KPC3595 cell line[42] was kindly provided by Ivonne Regel, Institute of Tumor Immunology, Medical Faculty of the Martin Luther University Halle-Wittenberg. It was cultivated in DMEM supplemented with 10% FBS and 1% penicillin-streptomycin.

Human NK-92 cells (ATCC) were cultivated in RPMI supplemented with 10% FBS, 1% penicillin-streptomycin, 1 U/µl IL-2, 0.1 ng/ml estradiol, 5 ng/ml testosterone (Thermo Fisher), or no hormones.

The human SU.86.86 cell line (ATCC) was cultivated in RPMI supplemented with 10% FBS and 1% penicillin–streptomycin. They were stably transduced with GFP to allow for cell identification in co-culture experiments using lentiviral transduction as previously described.[43] In brief, HEK293T cells (3.5×10^6) were transfected using Lipofectamine 3000 with the packaging plasmids psPax2, pMD2.G, and the pLVX-GFP lentiviral expression vector.[43] Virus titers were determined 48 hours post-infection by flow cytometry using a MACSQuantX Analyzer (Miltenyi Biotec). Transduction was performed at 5 MOI, and stably transduced cells were sorted for purity of >95% using a BD FACS Melody.

Animal housing and Allograft model

Healthy male and female C57BL/6JRj mice (Janvier Labs) at 12 weeks of age were used. They underwent a two-week acclimatization before entering the experiments. Mice for cell isolation and *in vitro* assays were euthanized at the Institute of Anatomy in Halle (n = 18/sex), while the allograft mouse study was conducted at the Helmholtz-Zentrum Dresden-Rossendorf. For tumor induction, 2×10^6 Panc02 cells in 100 µl of PBS were injected subcutaneously into the right leg (n = 5/sex); controls received only PBS (n = 3/sex). All procedures were performed under light anesthesia with 9% (v/v) desflurane (Baxter) in 30/10% (v/v) oxygen/air, and the mice were continuously warmed at 37 °C. The mice were monitored three times weekly for well-being and tumor growth via calipers, which were calculated by assuming a tri-axial ellipsoid with diameters a, b, and c using the formula $V = \pi/6 \times a \times b \times c$. All the mice were euthanized as soon as any of the tumors reached 300 mm^3 (four weeks). Tumours were collected and measured, and all organs were examined for metastases or abnormalities. Blood was sampled retroorbitally before tumor injection, and at 1, 2, and 4 weeks after injection, blood NK cells were analyzed by flow cytometry.[44] Euthanasia occurred in the morning during the light period by cervical dislocation.

NK cell isolation

Primary NK cells were isolated from the mouse spleen. Positive isolation was performed to obtain a pure NK cell suspension. The spleens were dissociated using the gentleMACS Dissociator (Miltenyi Biotec). The suspension was filtered, and after erythrocyte-lysis, stained with NK1.1 and CD3 antibodies (Miltenyi Biotec) as well as with DAPI live/dead staining and sorted by fluorescence-activated cell sorting (FACS) (BD FACS Aria II, BD Biosciences) for $DAPI^{-}NK1.1^{+}CD3^{-}$ cells.

Negative isolation was used to obtain unlabeled primary NK cells via magnetic-associated cell sorting (MACS) with an NK cell isolation kit (Miltenyi Biotec). Spleens were dissociated with a gentleMACS Dissociator (Miltenyi Biotec) and filtered, and non-NK cells were depleted by column separation, yielding an NK cell-enriched population. Isolation success was checked by flow cytometry. Based on the NK cell yields from each mouse, cell suspensions were prepared for use in killing assays.

NK cell killing assays

Killing assays were conducted by co-culturing primary negatively isolated NK cells with Panc02 and KPC3595 cells, as well as NK-92 cells with SU.86.86 cells. Target cells ($4500/cm^2$) were seeded in a well plate by FACS (BD FACS Melody, BD Biosciences) to ensure highly accurate cell numbers and, if applicable, GFP-positivity rates close to 100%. NK cells were added 24 hours post-seeding at 5:1 and 10:1 effector:target (E:T) ratio, and incubated for 24 h. An NK cell-only control was used to determine target cell-independent effects. The target cell only served as a growth control. All conditions were supplemented with 1 U/µl IL-2. NK input controls were analyzed to assess culture-dependent effects. Effector-focused killing assays were analyzed by flow cytometry and RNA sequencing. For target-focused killing assays, time-lapse imaging was carried out in an Incucyte S3 device (Sartorius) for 64 h using phase

contrast and GFP fluorescence with adherent cell-by-cell settings. Target cell segmentation and assessment of target viability were performed on phase contrast and GFP fluorescence using Incucyte S3 GUI 2022B rev2 software and the AI health module to determine the GFP intensities, the percentage of dead cells, and the number of living cells. The AI health classifications were validated by the addition of PI on termination of the experiment, confirming the percentage of dead cells.

Multicolor FISH (mFISH)

To determine the sex of the Panc02 cells, multicolor fluorescence in situ hybridization (mFISH) was performed.[45] Whole chromosome paints (wcps) for all 21 murine autosomes were obtained from the 21XMouse commercial kit (MetaSystems), as well as custom wcps for the X and Y chromosomes, which were generated in-house via glass-needle-based chromosome microdissection.[46] The chromosomal workup and FISH procedures, including denaturation, overnight hybridization, and washing procedures, were performed according to the manufacturer's instructions. At least 30 metaphases were documented and analyzed for each probe set on a Zeiss Axioplan microscope equipped with ISIS software (MetaSystems).

Flow cytometry

Flow cytometry was performed to control the success of the negative NK cell isolation, with isolated NK cells before and after co-culture, with peripheral blood from allografted mice, and with NK-92 cells before and after co-culture. In the framework of primary NK cell isolation, NK cells were stained with CD45, NK1.1, CD3, and PI for live/dead discrimination. Surface staining of primary mouse NK cells and allograft blood samples included CD45, NK1.1, CD3, NCR1, NKG2D, CD11b, CD27, and PI.[44] Surface staining of human NK-92 cells included Fc receptor block (Miltenyi Biotec), CD45, NKp30, NKp44, NKp46, NKG2D, NKG2A, and DAPI. Intracellular staining was exclusively performed with primary mouse NK cell cultures and included NK1.1, CD3, granzyme A and B, and a fixable live/dead stain. For this purpose, an Inside Stain Kit (Miltenyi Biotec) was used. Fluorescence minus one (FMO) controls were conducted for each antibody for which the fluorescence signal was not clearly separable into positive and negative signals or was poorly detectable. All antibodies are summarized in Supplementary Table T1.

The samples were analyzed via MACS Quant 16 (Miltenyi Biotec). The flow cytometry results were analyzed using FlowJo™ v10.10 Software (BD Life Sciences). Target cells were excluded by scatter, GFP fluorescence, and lack of CD45. Mouse primary NK cells were identified as $CD45^+CD3^-NK1.1^+$, NK-92 cells as $CD45^+$, and MFI values were assessed for every antibody.

RNA isolation

RNA from positively sorted NK cells was extracted using Dynabeads™ $Oligo(dT)_{25}$ RNA isolation kit (Thermo Fisher Scientific) according to the manufacturer's instructions. The quality of the RNA was assessed by agarose gel electrophoresis. Reverse transcription was performed with the Reverse Transcription System (Promega).

Total RNA from cell experiments or allograft tumors was isolated by TriZOL® (Thermo Fisher Scientific) according to the manufacturer's instructions.[47] For allograft tumors, the tissue was dispersed using an Ultra Turrax (IKA®). RNA integrity was determined with an RNA Pico kit using a Bioanalyzer 2100 (Agilent Technologies). Samples with RIN values between 8.5 and 10 were considered. The isolated RNA served as a template for RNA sequencing.

Real-time RT-PCR

RT-PCR was performed on a qTower^{3}G thermocycler (Analytik Jena), using primer sets listed in Supplementary Table T1 and the iQ SYBR Green Supermix (Bio-Rad). The PCR products were size separated by agarose gel electrophoresis.

RNA sequencing and differential gene expression

For library preparation, total RNA from cell culture (100 ng) or allograft samples (500 ng) was used. PolyA enrichment was performed using oligo dT beads, followed by strand-specific library construction with the Stranded mRNA Prep kit (Illumina) according to the manufacturer's instructions.[48] Paired-end sequencing was carried out on a NextSeq 1000 instrument (Illumina) with a P2 flow cell and a P200 cartridge (Illumina).

Low-quality reads and sequencing adapters were trimmed using Cutadapt (V4.4).[49] Reads were aligned to the murine genome (UCSC GRCm39) using HiSat2 (V2.2.1).[50] FeatureCounts (V2.0.6)[51] were used to summarize the gene-mapped reads. Ensembl (GRCm39.113) was used for annotation. Differential gene expression was determined by DeSeq2 (Galaxy V2.11.40.8+galaxy0) using the Galaxy-EU project (V25.0).

Gene set enrichment analysis (GSEA)

Log2 fold-changes in gene expression and their respective adjusted p-values (DESeq2) were used for gene ranking before GSEA was conducted[52,53]using the GSEA software (V4.3.2) and MSigDB (V2025.1Mm). Hallmarks, M2 chemical and genetic perturbations (M2cgp), M2 – canonical pathways Reactome and Biocarta, M7 – immunology gene set collections, NK cell sets by Cui et al. [54], and the UTX-KO gene set were analyzed. The UTX-KO gene set reflects the intersection of Kdm6a-KO, Kdm6a-Chip data, and cut&seq data according to Cheng et al. [25] The detailed composition of this gene set is provided in Supplementary Table T2.

Histological staining, imaging, and analysis

All tumors were partially fixed in 4% paraformaldehyde (Carl Roth GmbH), embedded in paraffin, and sectioned at 5 µm. After deparaffinization, the sections were stained with hematoxylin and eosin (Carl Roth GmbH) to assess the tumor structure, and NK1.1 immunohistochemistry was performed to identify infiltrating NK cells. Antigen retrieval was performed with citrate buffer (Zytomed Systems), followed by different blocking steps. The sections were incubated overnight with the primary antibody (Cell Signaling), then with a secondary antibody (Biozol), visualized with DAB (Vector Laboratories), and counterstained with Haemalaun (Carl Roth GmbH). The slides were scanned using the Panoramic Scan II (3D HISTECH Ltd.) and analyzed via QuPath (V0.3.2),[55] the tumor area was quantified, NK cells were annotated, and NK cell infiltration per area was calculated. The antibodies are listed in Supplementary Table T1.

Multiplex immunofluorescence imaging and cyclic staining (MICS)

High-dimensional multiplex immunofluorescence imaging was performed using the MACSima™ Imaging Platform (Miltenyi Biotec). FFPE tumor samples were processed as described above. Antigen retrieval was carried out by heat-induced epitope retrieval in Tris buffer (pH 9). Imaging was performed in five independent runs, each including two samples processed in parallel (one female and one male). Antibodies directly conjugated to fluorophores were applied sequentially over 12 imaging cycles, with fluorescence signal removal achieved by bleaching for 6 min between cycles. A complete list of antibodies and corresponding fluorophores is provided in Supplementary Table T1.

Prior to analysis, the images were pre-processed to exclude background fluorescence and residual signals from previous staining cycles. Image analysis, including cell segmentation, classification, gating, and TME subtyping, was performed using MACSima IQ View software (definitions provided in Supplementary Table T3). The mean fluorescence intensities were thresholded for all marker channels before scaled inverse hyperbolic sine transformation with a factor of 5.

Statistical analysis, bias control, and quality assurance

Unless stated otherwise, the visualization and statistical evaluation of the data were performed with GraphPad Prism (V10.4.2). Statistical tests and n-counts are indicated in the figure legends. The p-values are indicated in the graphs if they are smaller than 0.1 (90% confidence interval). The data are presented as median with individual values or standard error.

For multiplex immunofluorescence imaging, dimensionality reduction (t-SNE) and data visualization using violin plots were conducted in RStudio (2025.09.02 Build 418) using the Rtnse, ggplot2, and tidyverse packages. Statistical analyses were performed using the Wilcoxon test with Benjamini–Hochberg correction for multiple comparisons using Rstatix. The representative images shown depict male and female samples processed within the same imaging run.

There was no randomization of the animals. Potential confounding factors were minimized as much as possible by employing stress-reducing handling procedures, conducting all examinations and experiments at the same time of day, and including animals from all groups in each experiment. However, uncontrollable influences such as social rank, individual characteristics of the animals, cage position within the rack, or the order of treatments may have affected the results. Knowledge of group allocation was unavoidable during the experimental procedures and euthanasia, but following data collection, analyses were performed while the participants were blinded to group affiliation wherever possible to minimize bias. No statistical outliers were excluded from the analyses, but we excluded two mice (1 male, 1 female) used in the allograft model from the analysis of the RNA-seq data because the achieved RNA amount or quality was not sufficient for sequencing.

Use of large language models

During the preparation of this manuscript, we used a large language model (OpenAI ChatGPT, version GPT-4.1) to exclusively assist in the formulation and editing of certain text passages. All the content was critically reviewed and revised by the authors to ensure accuracy and compliance with academic standards.

Results

As sex-dependent differences have been reported for the incidence and mortality of PDAC (Figure 1),[39] as well as NK cell function, we aimed to elucidate the role of biological sex in NK cell-mediated immunity against PDAC-derived cell lines *in vivo* and *in vitro*.

Appropriate sex characterization of effector and target cells allows for unbiased sex-specific NK cell killing assays

To investigate potential sex-dependent differences in the interaction of NK cells with pancreatic cancer cells, we established a murine killing assay. For use as effector cells, we isolated primary NK cells from the

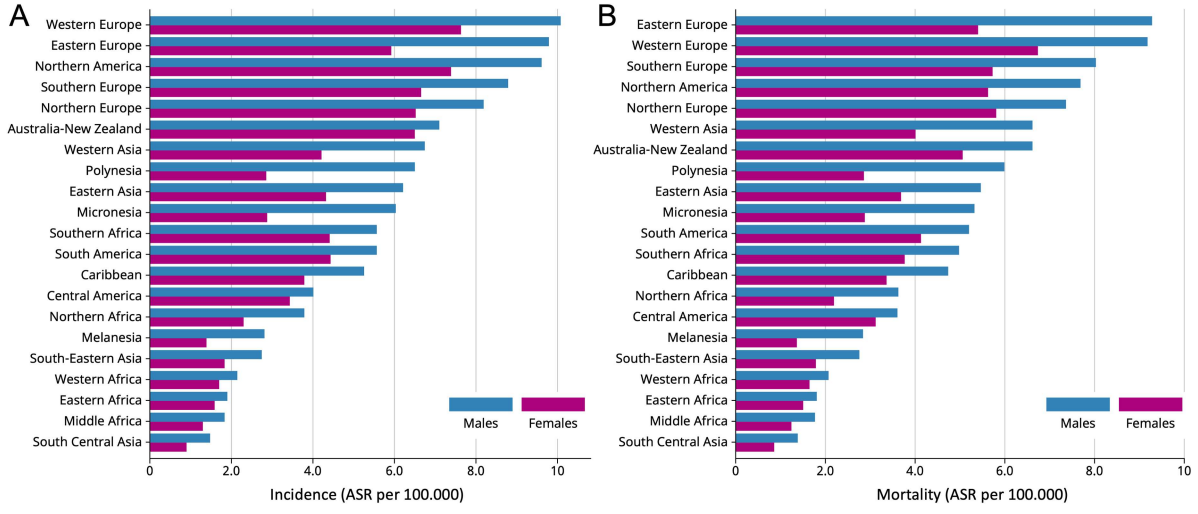


Figure 1. Sex differences in PDAC. (A) Worldwide incidence (age-standardized rate = ASR) for PDAC in 2022. (B) Worldwide mortality (ASR) for PDAC in 2022. From Cancer Today, IARC/WHO, https://gco.iarc.who.int (Globocan 2022, version 1.1, 08.02.2024), accessed 21 November 2025.[39]

spleens of male and female mice via negative isolation (Figure 2A). The isolated NK cells displayed purities around 85%–100% (Supplementary Figure S1), and their sex was confirmed via RNA-seq. As expected, we detected clear differences in the expression of X and Y chromosomal genes between male and female NK cells. *Kdm5d* and *Uty* were exclusively expressed in male NK cells, while *Kdm5c* and *Kdm6a* (UTX) were predominantly detected in female NK cells (Figure 2B). These *Kdm5/6* genes encode histone demethylases involved in epigenetic regulation, and notably, *Kdm5c* and *Kdm6a* are known to escape X-chromosome inactivation.[25]

As target cell line and PDAC model, we utilized the murine Panc02 cell line stably labeled with GFP, enabling rapid identification of tumor cells via downstream analyses. It is derived from chemically induced PDAC in a male C57BL/6 mouse.[57] As our study focused on sex-dependent interactions, we verified the sex chromosome status of Panc02 cells via mFISH and RNA-seq. mFISH identified hyperploidy of the Panc02 cells without any detection of the Y chromosome and a reduced number of X chromosomes (Figure 2C). Consistently, transcriptomic profiling revealed a significant reduction but not absence in Y-linked gene expression compared to freshly isolated male NK cells, along with low expression levels of X-linked genes (Supplementary Figure S2). In addition, the data were translated to the human genome,[45] which revealed notable similarities in chromosomal aberrations in Panc02 cells compared to human PDAC, with 16 of 25 regions (64%) being affected (Supplementary Table T4).[58] Therefore, the Panc02 cell line is a suitable PDAC model and was considered sex-neutral for the purposes of our study.

Naïve NK cells from males and females show a similar phenotype with subtle differences in cytokine response and histone methylation signatures

To elucidate the initial differences between male and female NK cells, we analyzed the phenotypes of freshly isolated NK cells via flow cytometry and RNA-seq. We found that freshly isolated NK cells displayed a naïve, non-activated phenotype, characterized by low expression of the maturity markers CD11b and CD27, and the activation markers NCR1 and NKG2D, as well as a low intracellular content of granzymes A and B (Figure 2D). We detected no significant differences between male and female NK cells (Figure 2D). In line with this, RNA-seq showed comparable expression levels of NK cell activation-associated transcripts in both sexes and minimal detection of markers for other immune cell lineages, confirming the isolation success (Figure 2E). The expression of *Ar* and *Esr1* did not differ markedly between freshly isolated male and female NK cells (Figure 2F, upper panel). Notably, *Esr2* was not expressed in primary mouse NK cells (Figure 2F, upper panel). The expression of *Ar* and *Esr1* was confirmed with RT-PCR in positively isolated NK cells (Supplementary Figure S3). The expression of their respective response genes differed only marginally between males and females (Figure 2F, lower panel).

Interestingly, we identified sex-dependent differences in the cytokine response and histone methylation signatures. In line with previous reports[25] and in agreement with the absence of *Kdm6a* in male NK cells (Figure 2B), *Kdm6a*-regulated genes were lower in male than in female NK cells (Figure 2G, upper panel). This includes genes linked to the repressive H3K27me3 histone mark,[56] which were expressed at lower levels in freshly isolated male NK cells compared to females (Figure 2G, lower panel). Although male and female NK cells showed similar baseline phenotypes (Figure 2D, E), IL-1A- and IL-1B-associated response genes were significantly upregulated in male NK cells, whereas response signatures for IFN-α, IL-12, and IL-15, which essentially enhance NK cell activity in a *Kdm6a*-dependent manner, were significantly downregulated (Figure 2H and Supplementary Table T5).[25,54] These findings are consistent with previous reports showing reduced IL-12 and IL-15 responsiveness in males.[25] This finding indicates that female NK cells are epigenetically primed for enhanced killing capacity.

NK cell maturation in co-culture with Panc02 is accompanied by immune evasion-driven functional downregulation

To investigate tumor cell-related changes in the NK cell phenotype, we analyzed the killing assays by flow cytometry and RNA-seq. We used freshly isolated NK cells, as well as co-cultures with Panc02 and NK only controls (Figure 3A$_1$), which were incubated for 24 h with IL-2. A clustering of NK cells around Panc02 cells was observed after 4 h of co-culture (Figure 3A$_2$). We found that IL-2 could activate NK cells

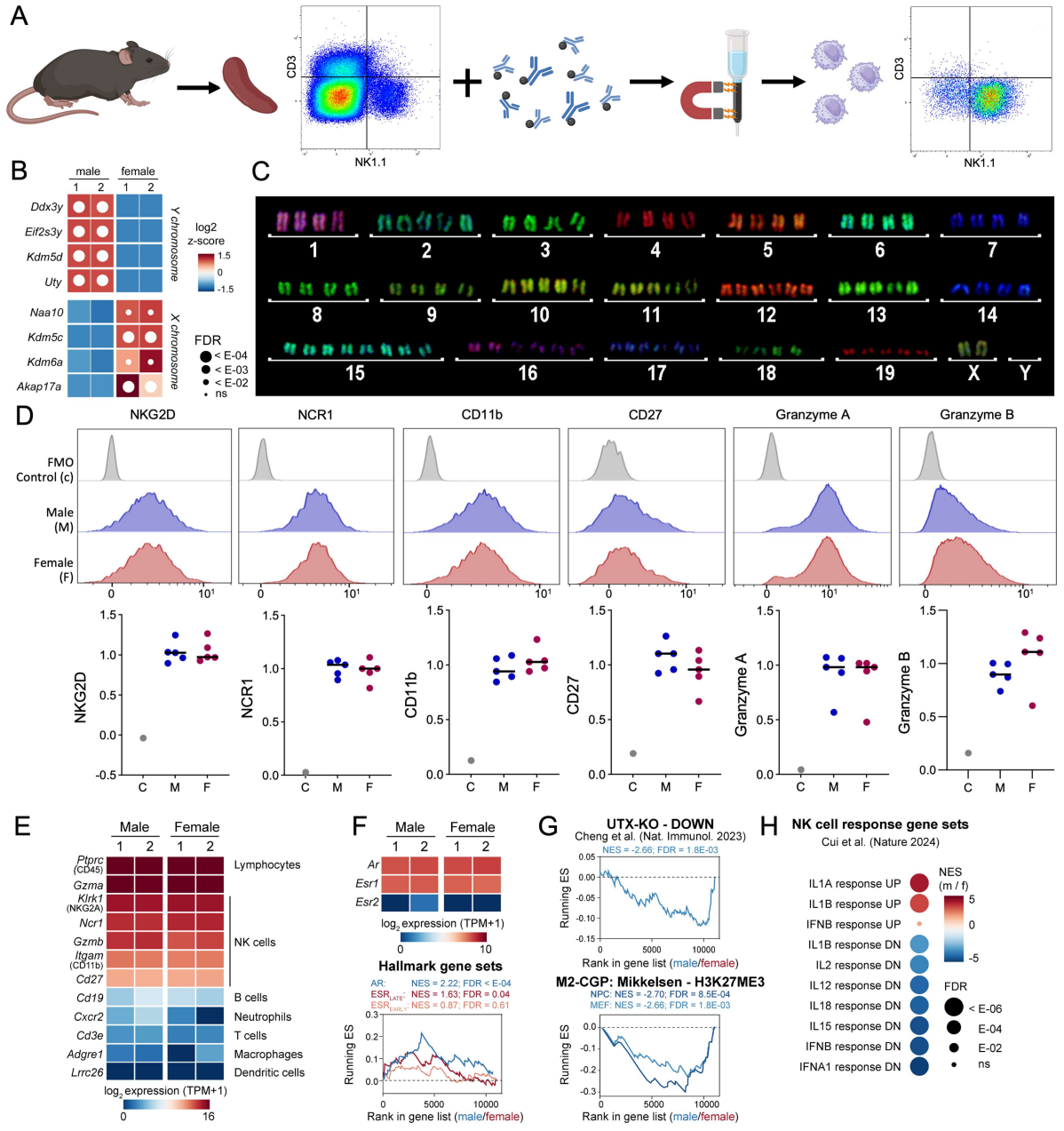


Figure 2. Phenotype of primary NK cells after isolation. (A) Schematic illustration of primary NK cell isolation from mouse spleens using magnetic-associated cell sorting (MACS). Spleens were dissociated and filtered to yield single-cell suspensions containing 2%–7% $CD3^{-}NK1.1^{+}$ cells. Non-NK cells were magnetically labeled and depleted via column separation, resulting in an NK cell-enriched population of 85%–100% purity. (B) Heatmap of X and Y chromosomal genes confirming the sex of freshly isolated NK cells. RNA-seq-derived reads (TPM) are indicated upon z-transformation. FDR was determined by calculating differential gene expression using DESeq2; n = 2/sex. (C) Karyogram of Panc02 cells. The karyotype 85~92<4n>,XX,-Y,-Y,+2,+10,+11,+12×2,+15×4,+16×2,+17×2,+19 indicates hyperploidy but loss of the Y chromosome. (D) Histograms (upper panels) and median fluorescence intensity (MFI, lower panels) values for NKG2D, NCR1, CD11b, CD27, granzyme A, and granzyme B in freshly isolated NK cells. The data were normalized to daily input, and an unpaired t-test displayed no significant differences between males and females, n = 5/sex. (E) Heatmap indicates $\log_2$ expression (TPM+1) of the indicated marker genes for different immune cell lineages. Low reads for CD3e, Adgre1 (F4/80) and Lrrc26 confirmed the purity of the NK cell isolates. n = 2/sex. (F) Heatmap indicates $\log_2$ expression (TPM+1) of *Ar*, *Esr1* and *Esr2* (upper panel). The enrichment plots indicate enhanced expression of the indicated hormone-related hallmark gene sets in male vs. female freshly isolated NK cells (lower panel), n = 2/sex. (G, H) GSEA indicates a downregulation of Kmd6a-regulated or H3K27-trimethylated genes (G) and alterations in NK cell immune response gene sets (H) in freshly isolated male NK cells. GSEA was performed using fold changes in gene expression between male and female NK cells, with gene ranking based on the significance of alterations, for UTX-KO (G, upper panel),[25] m2-CGP (G, lower panel),[56] and m7-NK cell collections (H).[54]

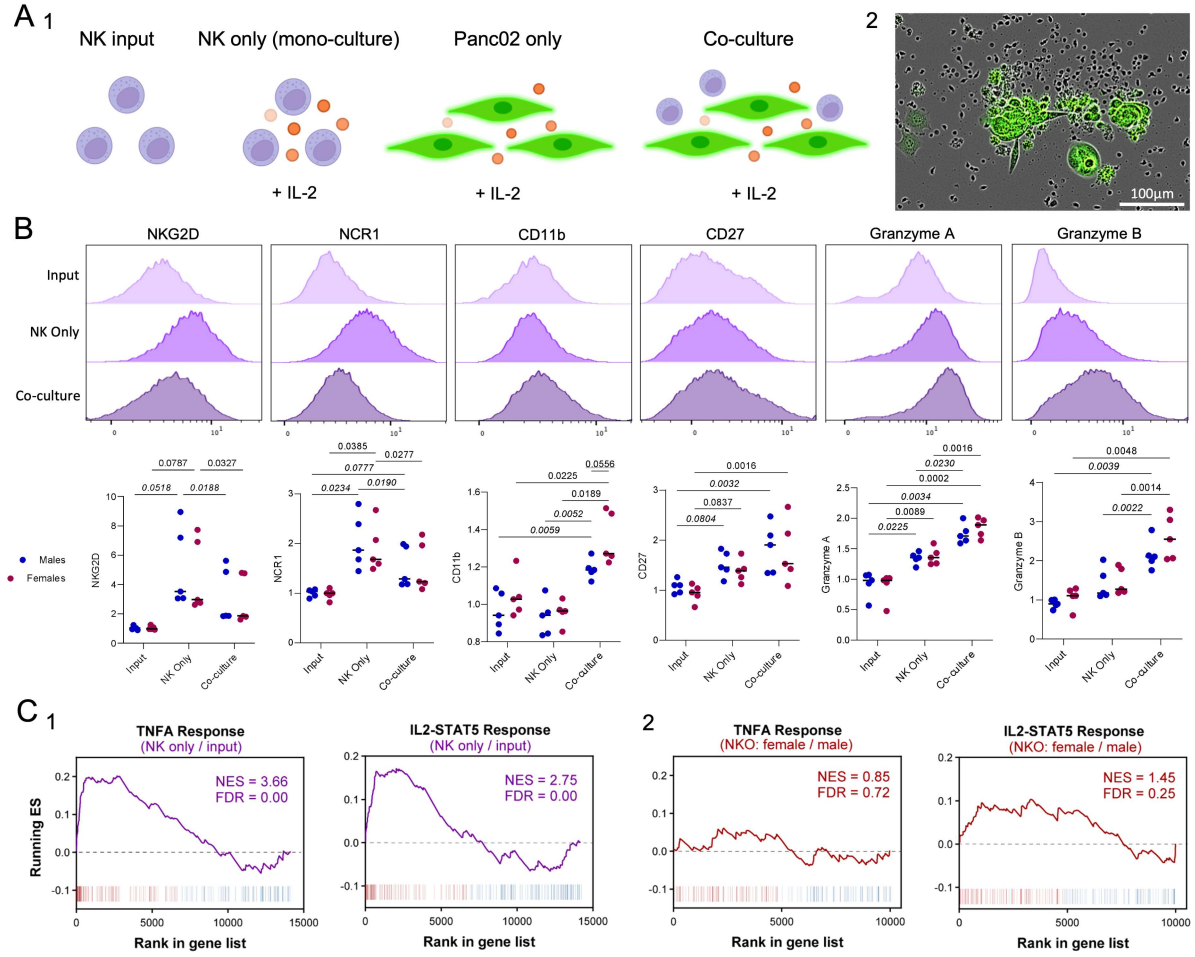


Figure 3. NK cell maturation is associated with functional downregulation in co-culture with Panc02. (A) Schematic illustration of the experimental conditions for the killing assays (1). GFP-overlayed brightfield image indicates that NK cells cluster around GFP-Panc02 cells after 4 h (2). (B) Histograms (upper panels) and median fluorescence intensity (MFI, lower panels) values of NKG2D, NCR1, CD11b, CD27, granzyme A, and granzyme B in NK cells following isolation (input), 24 h NK mono-culture (NK only), and 24 h Panc02 co-culture. Data presented as median and normalized to daily input. Statistical significance was assessed by two-way ANOVA with Tukey's post hoc test; *p*-values: italics (male), regular (female), underlined (sex differences, same condition), $n = 5$/sex. (C) GSEA of the TNFA response and the IL2-STAT5 response based on ranked gene lists derived from log2 fold changes between mono-culture vs. input in both sexes (1) and female vs. male in mono-cultures (2). $n = 2$/sex.

in a target cell-independent manner, since the presentation of the activating surface receptors NCR1 and NKG2D increased compared to fresh isolates (input) (Figure 3B). In agreement, IL2-STAT5 and TNF-α signaling response genes increased sex-independently in the NK-only condition (Figure 3C and Supplementary Table T6). Importantly, the increases in NCR1 and NKG2D were less pronounced upon co-culture with Panc02 cells, suggesting a target cell-dependent reduction in NK cell activity. CD11b was more affected by co-culture with Panc02 cells than by IL-2 alone, indicating enhanced NK cell maturation in the presence of pancreatic cancer cells[59] (Figure 3B). The levels of the cytotoxicity-associated maturation marker CD27[59] and intracellular granzymes A and B levels were upregulated with increasing IL-2 in mono-culture and further increased during co-culture (Figure 3B). The only statistically significant difference between the sexes was found for CD11b in the co-culture setup, which was slightly higher for females compared to males, suggesting a further maturation process in female NK cells in the presence of Panc02 cells[59] (Figure 3B).

In summary, NK cells are activated and mature in both mono- and co-culture; however, their activation is relatively suppressed by Panc02 cells, with female NK cells showing a slightly higher degree of maturation.

Female NK cells display enhanced immune responses and cytotoxic capacity in co-culture with Panc02

To further examine sex-dependent differences in NK cell-tumor cell interactions, we examined the RNA-seq data from the killing assays. To distinguish NK cells from Panc02 tumor cells in co-cultures, we derived marker genes from the respective mono-cultures that clearly segregated the two populations (Supplementary Figure S4A). We detected a significant reduction in the number of tumor cells and, correspondingly, a relative enhancement of NK cells in female co-cultures (Figure 4A). Among the genes whose expression was most highly upregulated in female co-cultures, a variety of interferon-stimulated genes (ISGs), indicating an enhanced immune response were identified (Figure 4A). This phenomenon was conserved in another PDAC cell line (KPC3595), although immune responses were less prominent in these cells, likely due to the additional alteration of IL-10 response genes suggested to act as immune suppressors (Supplementary Figure S4B). Consistently, GSEA revealed enrichment of hallmark immune-response gene sets in female co-cultures of both cell lines (Figure 4B and Supplementary Figure S4C), while proliferative programs such as MYC and E2F targets exhibited reduced normalized enrichment scores (NES) (Figure 4B and Supplementary Tables T6+T7). This pattern was corroborated by Reactome, Biocarta, and Gene Ontology Biological Process (GOBP) analyses, which indicated decreased proliferation and enhanced immune activity (Figure 4C+D, Supplementary Figure S4D+E, and Supplementary Tables T7–T9).

Our analyses suggest that male and female NK cells are activated by different mechanisms. Pathways such as the Reactome: Immunoregulatory interactions between lymphoid and a non-lymphoid cell, Biocarta NKT pathway, and NK cell pathway were enriched exclusively in female co-cultures (Figure 4B+C, and Supplementary Figure S4D+E). In contrast, no sex-dependent enrichment of immune-related gene sets was observed in mono-cultures or freshly isolated NK cells (Figure 4B+C, and Supplementary Figure S4D+E). In female co-cultures, expression, interaction, and signaling of DAP12 (*Tyrobp*) were among the most prominently upregulated reactome pathway sets (Figure 4C and Supplementary Table T7). DAP12 is an ITAM-containing adaptor that pairs with NK cell receptors such as NKG2D (*Klrk1*) to modulate downstream signaling.[60] Recent studies have shown that CARs containing NKG2D and DAP12 domains reduce cytokine production but enhance tumor cell killing, highlighting the importance of this receptor in NK cell cytotoxicity.[61] Consistent with this, distinct response gene sets were enriched in female versus male co-cultures (Figure 4C and Supplementary Table T7). Female-enriched response gene sets were exclusively observed in co-cultures, indicating induction by target-cell contact (Figure 4D and Supplementary Table T10). In male cultures, upregulated pathways associated with the cytokine response were likely induced by IL-2, as these programs were elevated in both mono- and co-cultures (Figure 4D and Supplementary Table T11).

In conclusion, female NK cells display contact-dependent activation mediated predominantly by the IL-1α/β- and IFN-β-associated response pathways in co-culture with two independent cell lines, while males show IL-2-dependent activation. These results clearly indicate differential anti-tumor immune responses in our study.

Female NK cells kill Panc02 cells more efficiently than male NK cells in a sex hormone-independent manner

In order to quantify the killing efficacy of NK cells in a sex-dependent manner, killing assays were analyzed via time-lapse imaging with the Incucyte S3 device (Sartorius) over 64 hours. The GFP signal was assessed to model Panc02 proliferation curves (Figure 5A). We observed a slower proliferation of Panc02 cells in the presence of NK cells, which was more pronounced at higher E:T ratios, indicating that NK cells were killed in a dose-dependent manner (Figure $5A_2$). In addition, we found that Panc02 cell proliferation was significantly lower when co-cultured with female NK cells compared to male NK cells in both E:T ratios (Figure $5A_{2+3}$). The differences in the GFP proliferation curves between males and females were confirmed by analyzing AI-based single-cell segmentation of Panc02 cells (Live Count, Supplementary Figure S5A). In line with this, the AI-assessed percentage of dead Panc02 cells increased with higher E:T ratios and was

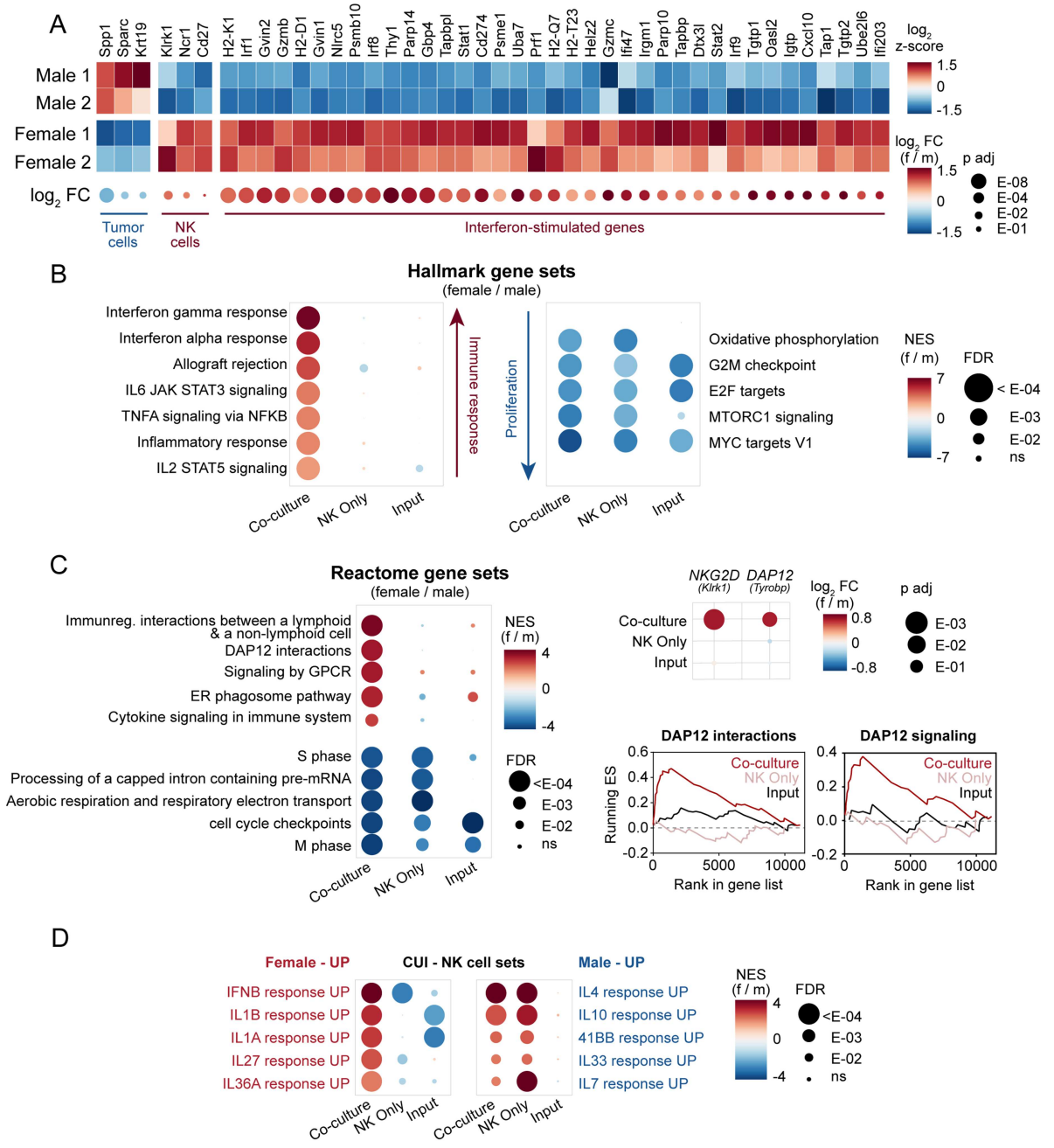


Figure 4. Female and male NK cells respond with different transcriptomic signatures. RNA-seq was performed with freshly isolated NK cells (Input), IL-2-activated NK mono-culture ("NK only"), and NK cell-Panc02 co-cultures derived from the same animal, $n = 2$/sex. (A) Heatmap showing tumor cell markers, NK cell markers, and interferon-stimulated genes (ISGs) derived from RNA-sequencing of female and male co-cultures after z-transformation. Log2-fold changes in gene expression and statistical significance are depicted as a bubble plot below the heatmap. (B–D) GSEA was performed using log2-fold changes in female versus male gene expression, with statistical significance used for gene ranking. Enrichment was evaluated for the Hallmark (B), Reactome (C), and NK cell signature collections associated with cytokine responses described by Cui et al. (53) (D) in NK-Panc02 co-cultures, IL-2-activated NK cells (mono-cultures), or freshly isolated NK cells (input). The complete GSEA results are provided in Supplementary Tables T6–T9. DAP12 interactions and DAP12 signaling are presented as enrichment plots (C, lower right).

consistently larger in female co-cultures (Figure 5B). This indicates that female NK cells not only restrict growth but also increase the destruction of target cells by approximately 50% compared to males in our study. This finding was validated by PI staining at the endpoint of the experiment, confirming the AI-based analyses (Supplementary Figure S5B). In agreement with our flow cytometry and RNA-seq data,

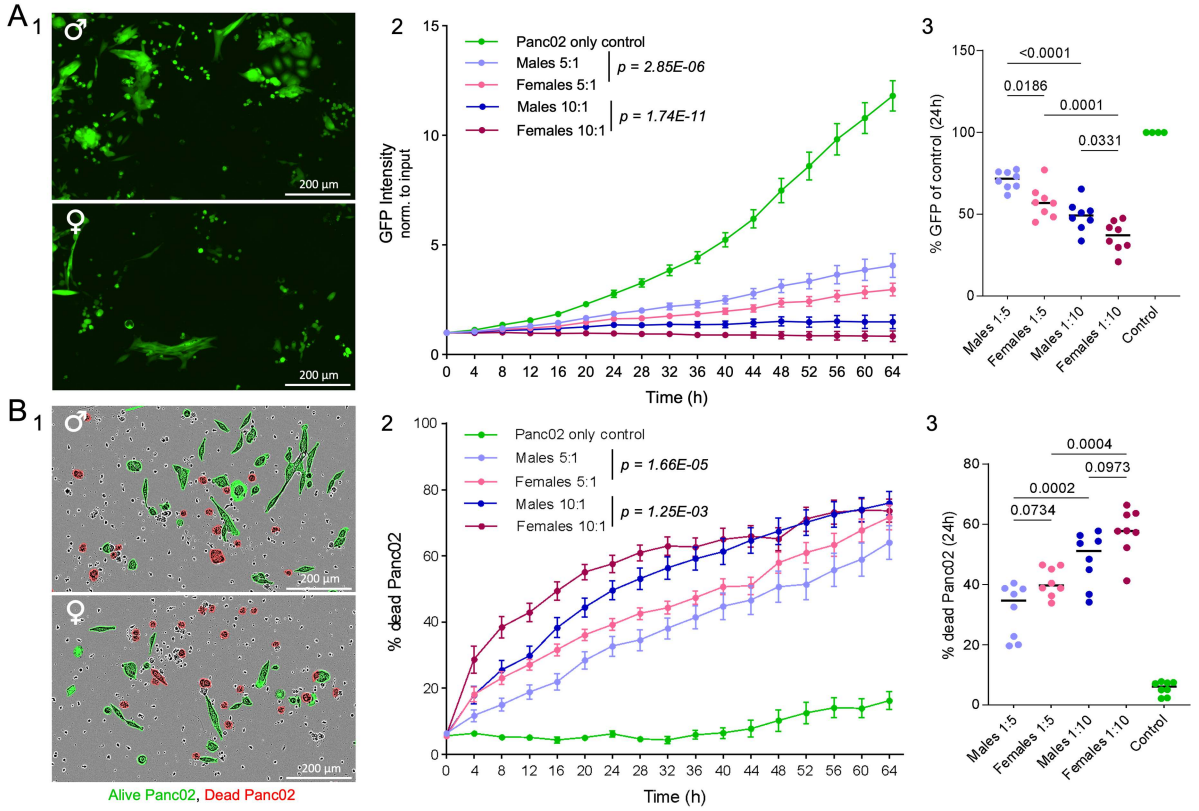


Figure 5. Female NK cells display enhanced killing activity compared to males, assesed via time-lapse imaging of NK cell co-cultures with GFP-labeled Panc02 cells. Co-cultures at the indicated effector-to-target ratios were monitored in an Incucyte S3 device in the presence of IL-2. GFP expression was used to distinguish target cells from effector cells and to model proliferation curves, while single-cell segmentation was performed with the AI Health Module based on brightfield images to determine living and dead tumor cells. (A) Representative GFP fluorescence images (1) and growth curves generated by analyzing the GFP intensity relative to that at baseline (2). The 24 h time point highlights differences between groups and is normalized to the Panc02-only control (3). (B) Segmentation mask-overlayed brightfield images indicate alive Panc02 in green and dead Panc02 in red (1). The growth curves represent the percentage of dead Panc02 cells (2). The 24 h time point highlights differences between groups (3). The growth curves are presented as the median ± standard error. Statistical significance was assessed by Chi-square test[62] and one-way ANOVA with Tukey's post hoc test, *n* = 8/sex with 7 technical replicates.

these findings indicate that female NK cells show higher anti-tumor immune responses and killing capacities against Panc02 cells than male NK cells.

In order to investigate the potential effect of sex hormones on NK cell cytotoxicity, we performed co-culture experiments using the human NK-92 cell line and the PDAC cell line SU.86.86.[63] NK-92 cells were stimulated for at least one week with physiological concentrations of estradiol and testosterone, after which their cytotoxic efficacy against SU.86.86 cells was assessed. This approach mirrors our experiments with primary NK cells, which were exposed to hormones until isolation, but were not subjected to hormonal influence during the actual killing assay. Although we detected initial differences in NK cell activity following hormone stimulation, these effects dissipated within 24 hours in the presence of SU.86.86 and in the absence of hormones (Supplementary Figure S6A) and had no impact on the cytotoxic efficacy (Supplementary Figure S6B). This finding is in line with the findings from primary NK cell co-cultures, where we observed an initially higher androgen response in freshly isolated male NK cells, which subsided in co-culture with Panc02 cells, while estrogen responses differed only insignificantly (Supplementary Figure S6C). Additionally, the experiments showed that the initial effect triggered by sex hormones vanished within 24 h, which is why hormones presumably had no influence on the *in vitro* killing assays. Given that the effect of sex hormone stimulation prior to the killing assay did not affect the NK-92 cytotoxicity, while enhanced NK immunity persisted in

female primary NK cells after isolation, these findings strongly suggest that the observed effect is independent of sex hormones.

Slower growth of female allograft tumors is associated with higher NK cell infiltration

To examine the interaction of NK cells with pancreatic cancer cells *in vivo*, an allograft mouse model was established. For that purpose, Panc02 cells were injected into the right leg of male and female mice (Figure 6A). Multiple blood samples were collected during tumor development and analyzed via flow cytometry. A decrease in peripheral NK cells was evident in allograft females. No significant differences in quantity, activity, or maturation were observed between the groups (Supplementary Figure S7).

The tumor growth was quantified three times per week. We measured faster tumor growth in males than in females (average doubling time: 20.8 d in males vs. 37.4 d in females; final volume: 196.0 vs. 67.6 mm^3, Figure 6B). Histological H.E. staining of the tumors revealed a homogeneous tumor composition of Panc02 cells with individual infiltration into different subcutaneous tissues (Supplementary Figure S8A–D). Immunohistochemistry of NK1.1 revealed infiltration of NK cells into the tumor tissues, especially at the tumor margin (Figure $6C_1$, Supplementary Figure S8E). We detected more NK cells infiltrating female tumors than male tumors (Figure $6C_{1+2}$). In line with these findings, we observed more NK cell-associated transcripts and fewer tumor cell-associated transcripts in females compared to males (Figure $6C_3$). Furthermore, tumor size is negatively correlated with NK cell infiltration, underscoring the connection between greater NK cell presence in females and reduced tumor burden. (Figure 6D). Our findings suggest that reduced Panc02 tumor growth in females in our study is at least partly associated with the NK cell function.

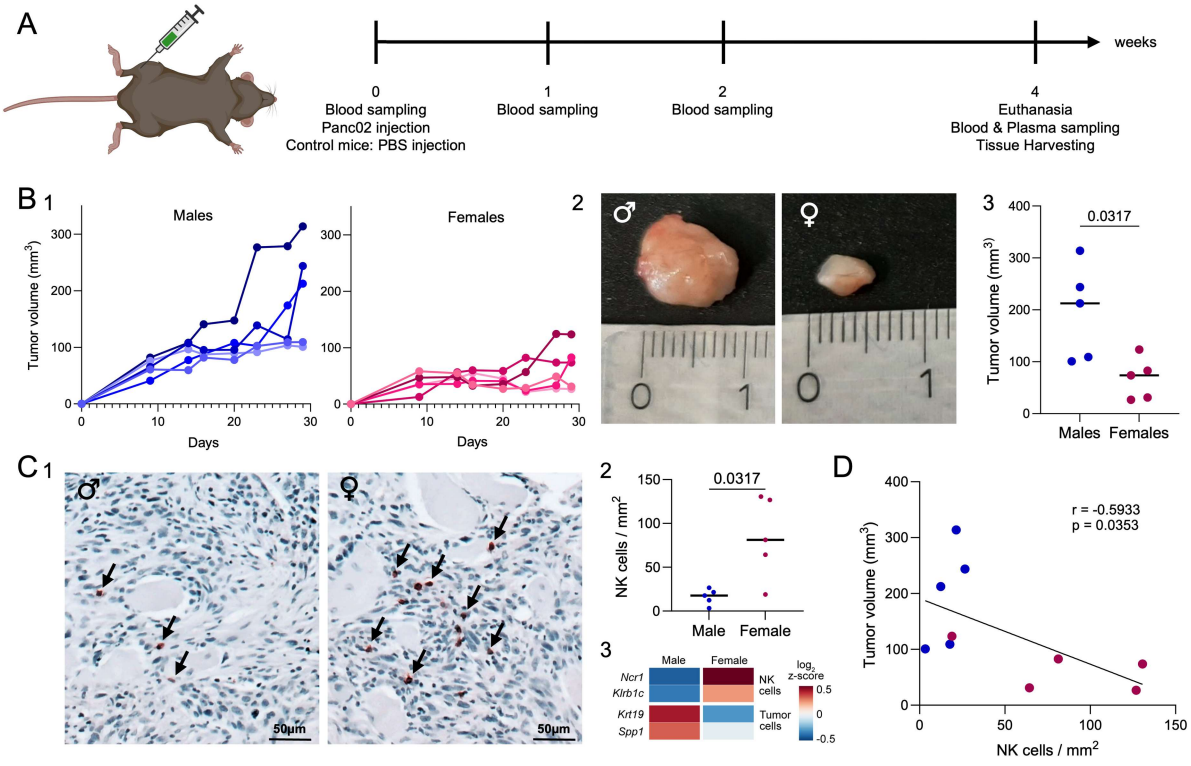


Figure 6. Panc02 allograft model shows reduced tumor growth but enhanced NK cell infiltration in female mice. (A) Schematic illustration of the experimental design. (B) Growth curves of male and female tumors (1) and representative images of excised tumors (2). Tumor volume at termination (3). Statistical significance was determined by Mann–Whitney U test, n = 5/sex. (C) Representative images for immunohistochemical staining for NK1.1; males vs. females; arrows indicate infiltrating NK cells (1). Scatter dot plot showing NK cell counts per area; statistical significance was determined by Mann–Whitney U test (2). Heatmap of RNA-seq expression for tumor and NK cell markers (3), n = 5/sex. (D) Correlation between NK cell infiltration and tumor size, simple linear regression and one-tailed Pearson correlation, n = 5/sex.

Female and male allograft tumors differ in their anti-tumor immune programs and their TME composition

To investigate sex-dependent differences in allograft tumors in greater detail, we combined RNA sequencing with high-dimensional multiplex imaging and cyclic staining (MICS) on the MACSima platform and analyzed more than 260,000 single cells based on functional gating strategies. By integrating differential gene expression analysis and gene set enrichment analysis (GSEA) with a MICS panel of 18 markers for tumor, immune, and stromal cells, we were able to characterize the molecular profiles and spatial organization of these tumors at the protein level, which is especially important in cancers with highly reactive stroma, such as PDAC (Figure 7A).

Although both male and female tumors are derived from the same PDAC cell line, their architecture and microenvironment differ markedly. Male tumors have a dense, compact structure with a higher abundance of immune-suppressive cell types and a stiffer stroma. In contrast, female tumors featured a looser connective tissue and increased infiltration by immune cells that are involved in anti-tumor responses (Figure 7B). Marker-based cell type profiling of the RNA-seq data revealed that female tumors contained greater proportions of NK cells, MDSC-like cells, T cells, non-M2 tumor-associated macrophages (TAMs), endothelial cells, and inflammatory cancer-associated fibroblasts (iCAFs) (Figure 7C). Correspondingly, female tumors contained fewer tumor cells, myofibroblastic CAFs (myCAFs), neutrophils, and M2-like TAMs compared to male tumors. This composition was confirmed by spatial MICS quantification, which showed non-M2 TAMs as the most prominent immune cell type in female tumors. In males, neutrophils and myCAFs were predominant (Figure 7D). Overall, myCAFs and iCAFs together accounted for 23% of intratumoral cells, while M2-like TAMs made up 18% of intratumoral cells, closely mirroring the cellular landscape of human PDAC with highly reactive stroma and predominant M2-TAM infiltration.

At the transcriptional level, female tumors exhibited significant enrichment of interferon response and allograft rejection gene sets, highlighting increased immune activation. In contrast, male tumors presented upregulated proliferation pathways, including E2F and MYC targets, the G2M checkpoint, and mTORC1 signaling (Figure 7E, Supplementary Table T12). These differences were consistent with co-culture experiments, as the same hallmark and response gene sets were upregulated *in vitro* and *in vivo* (Figure 4B, Supplementary Figure S9A, B).[54] Increased proliferative activity in male tumors was also supported by increased Ki67 expression (Figure 7F). The heightened immune activation in female tumors was further evidenced by an enrichment of NK cell response genes and markers of cytotoxic activity, such as elevated granzyme B expression in infiltrating T and NK cells (Figure 7G). Thereby, enhanced NK cell response gene sets in the allograft mirrored the findings from the co-culture setting (Supplementary Figure 9B, Supplementary Table T13).[54] Consistently, MHC-I and PD-L1 expression was globally higher in female tumors and their immune compartments, reflecting stronger immune engagement (Figure 7H).

Notably, the analyses revealed that neutrophils and MDSC-like cells in female tumors presented higher MHC-I to PD-L1 ratios, while the same cells in male tumors tended to express more PD-L1, implying more pronounced immune suppression in males (Figure 7H). CD44 expression also followed a sex-dependent pattern: it was increased in NK cells in female tumors but was associated with proliferative and immune-evasive tumor cell populations in males (Figure 7I). Differential expression analyses of genes with effector functions further supported these observations; for example, females upregulated CXCL11 and IGFBP6, while males showed higher levels of Cxcl2, Cd24a, and Alpl, which was consistent with enhanced neutrophil recruitment, macrophage evasion, and greater extracellular matrix stiffness (Figure 7J).

Fibroblast distribution also differed by sex. In males, SMA^+ myCAFs were abundant and associated with dense, poorly vascularized tumors, while in females, myCAFs were mainly found around $CD31^+$ vessels, and most tumor-infiltrating fibroblasts consisted of iCAFs embedded in a looser, immune-permissive collagen matrix (Figure 7K).

Minor yet consistent alterations in androgen and estrogen response programs were observed in the tumor tissues, potentially contributing to differences in tumor growth and immune regulation and exceeding those observed in mono- and co-culture settings (Figure 7L, Supplementary Figure S6C).

Beyond immune and stromal programs, female tumors consistently exhibited globally reduced levels of H3K27me3, especially in tumor, T, and NK cells, along with decreased SUZ12 expression and reduced

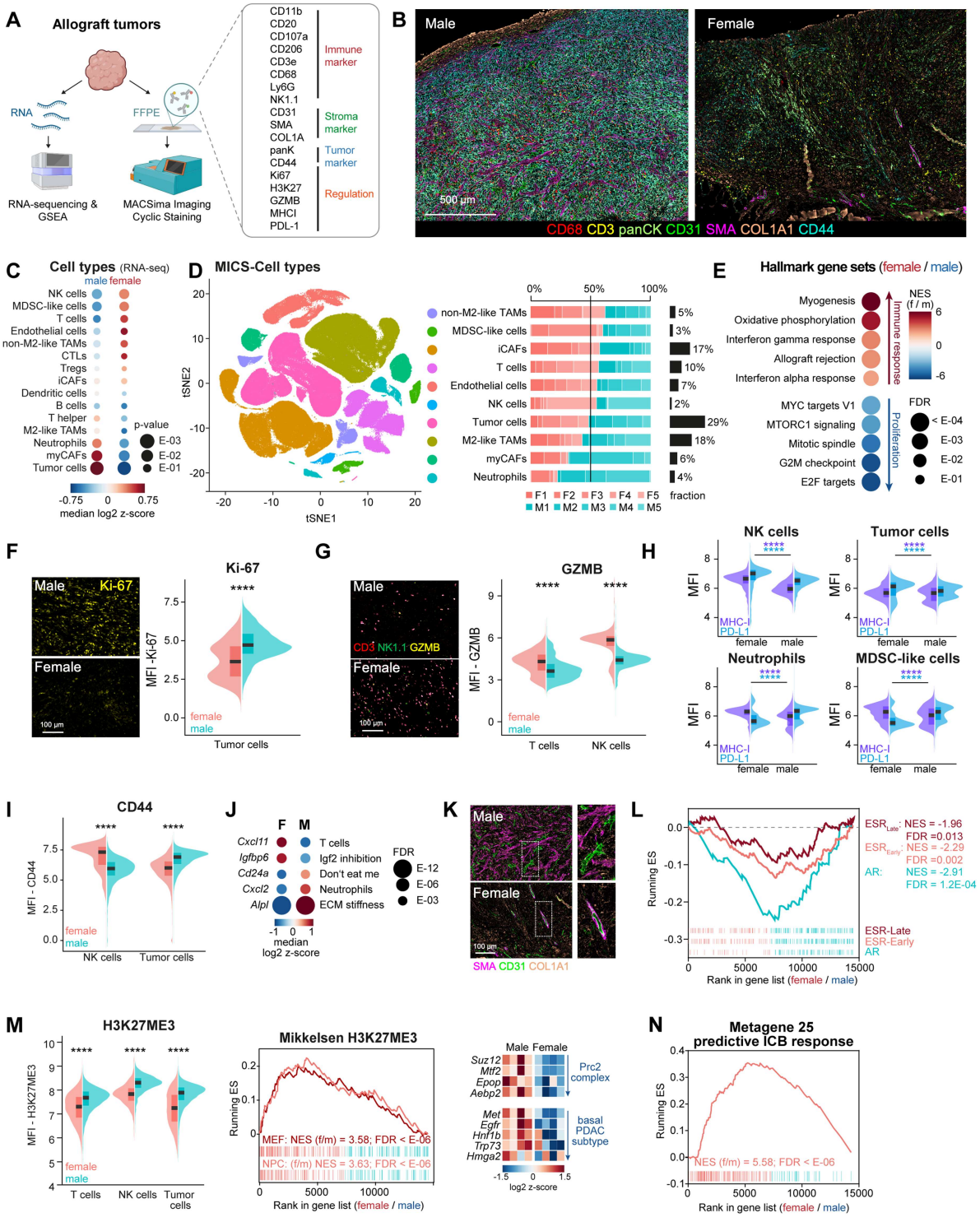


Figure 7. Female allograft tumors activate enhanced immune-response programs. (A) Scheme depicting the allograft tumor analyses. RNA-sequencing with GSEA was combined with multiplex imaging and cyclic staining (MICS) using the indicated panel from n = 5/sex (MICS) and n = 4/sex (RNA-seq). (B) Representative multiplex images of the indicated fluorescent markers from parallel experiments demonstrated sex-dependent tumor, immune, and stroma architecture. (C) Marker-based cell type profiling of male and female tumors was performed using RNA-seq. The marker panels used are summarized in Supplementary Table T3. N = 4/sex. (D) MICS analyses of 260,041 cells from 10 specimens (female: F1–F5; male: M1–M5) were performed using the multiplex panel indicated in (A). The cell types were annotated based on protein marker expression and gating strategies as described in Supplementary Table T3. The cell types are displayed as a t-SNE plot (left), while the population frequencies and their distribution among specimens are shown as bar charts (right). Analyses (right) were restricted to tumor regions, excluding the surrounding stroma. (E) GSEA was performed using log2

(*Caption on next page*)

fold changes in gene expression from tumors of female versus male mice, with statistical significance used for gene ranking. Bubble charts indicate the NES and significance for the top deregulated hallmark gene sets. The complete analyses are provided in Supplementary Table T11. (F–I) Sex-specific differences derived from the MICS analyses are shown for Ki-67 expression (F), the NK/cytotoxic T cells immune compartment (G), MHC-I and PD-L1 expression in different cell types (H), and CD44 expression (I). Representative images from the same imaging run are shown for Ki-67 and granzyme B (GZMB) (left panels). Violin plots indicate signal distributions for the indicated cell types (F, G, I) or sexes (H). Statistical significance was determined by the Wilcoxon test with Benjamini–Hochberg correction. (J) Bubble chart indicating the median sex-specific alterations for selected top-deregulated genes involved in the immune response and ECM organization. (K) MICS-derived images of the indicated marker genes indicate sex-specific differences in myCAF localization and endothelial cell organization. (L) Enrichment plots illustrate changes in the indicated hallmark hormone response gene sets between female and male tumors. (M) H3K27me3 alterations in female versus male tumors. H3K27 methylation levels derived from MICS are shown as violin plots for the cell type exhibiting the strongest alterations (left). Changes in H3K27me3-responsive genes are displayed as enrichment plots for the C2GCP MEF and NPC gene sets (middle).[56] Alterations in PRC2 subunits and basal PDAC subtype markers are shown as a heatmap (right). (N) GSEA of female versus male tumors indicates enhanced expression of genes predicting the PD-1 response.[64]

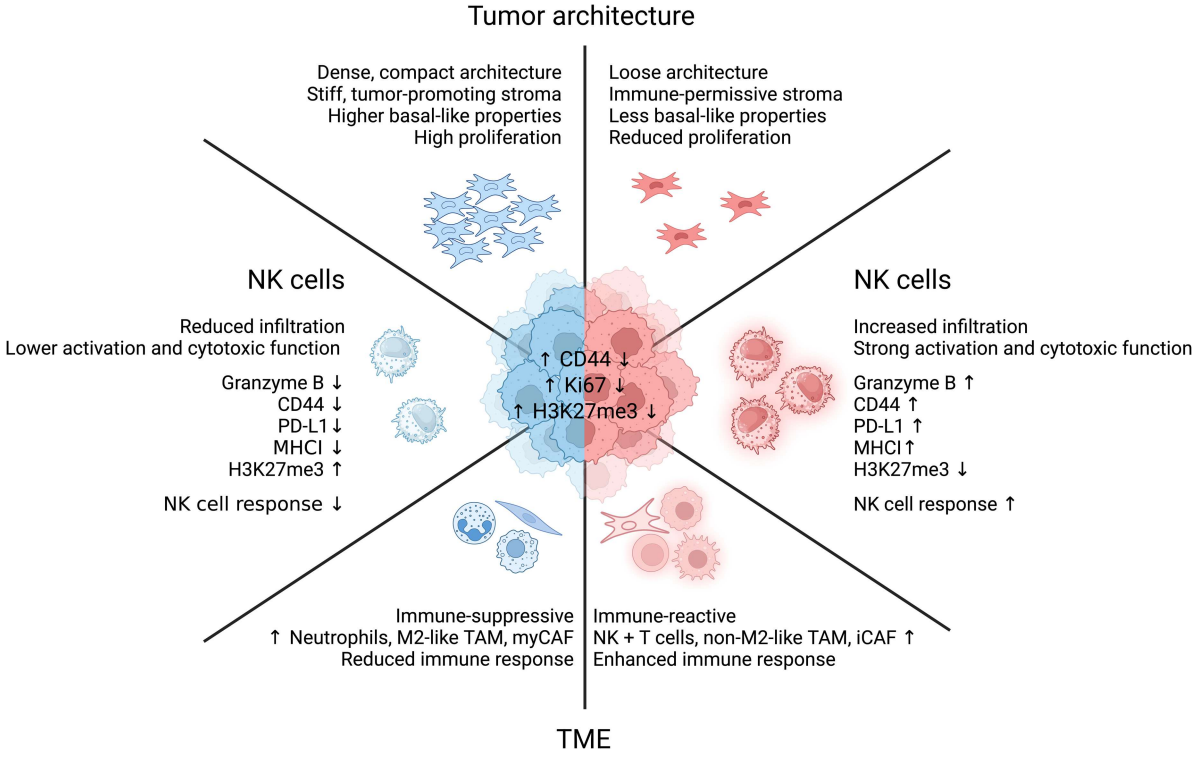


Figure 8. Composition of male and female allograft tumors. Overview of the different tumor compositions in male and female allografts, with a special focus on NK cells. Female tumors exhibit a loose tumor structure with infiltration of immune-reactive cells along with highly functional NK cells. Male tumors are dense, containing immune-suppressive cells and downregulated NK cells. The graphic is based on the results from Figure 7 and Supplementary Figure S9; blue = male, red = female. Created in BioRender. X, E. (2026) https://BioRender.com/n1vj20s.

PRC2 activity (Figure 7M). This epigenetic state correlated with increased expression of H3K27me3-responsive and Kdm6a-regulated genes and with the dampening of basal-like programs involving MET and EGFR (Figure 7M and Supplementary Figure S9).[25,65-67] Interestingly, GSEA indicates upregulation of genes associated with immune checkpoint blockade sensitivity, suggesting that female tumors may particularly benefit from immunotherapy (Figure 7N, Supplementary Table T14).[64]

Overall, these results show that female tumors are less proliferative and more immunologically active, characterized by an NK cell-enriched, immune-permissive stroma driven by a combination of epigenetic and, to a certain extent, hormone-dependent cues. In contrast, male tumors are more proliferative, aggressive, and immunosuppressive, underscoring sex as a key factor in tumor biology and therapy response (Figure 8). Although NK cells made up only a small fraction of the tumor microenvironment, they showed some of the most pronounced sex-specific differences. Both their function and epigenetic

state, particularly with respect to H3K27me3 methylation, differ markedly between females and males, indicating a strong influence on the immune-reactive tumor phenotype.

Taken together, the findings from our study indicate that NK cell-related immune responses against PDAC are stronger in females than in males in both *in vitro* and *in vivo* mouse models. Initial differences in NK cell cytotoxicity appear to be primarily driven by epigenetic mechanisms that operate independently of sex hormones.

Discussion

It is generally known that PDAC incidence and survival are worse for men than for women and that women are able to mount a stronger immune response than men.[13,37] NK cells are key mediators of antitumor immunity and are known to exhibit sex-specific differences;[14,68] however, whether this extends to PDAC, potentially contributing to divergent PDAC pathologies between the sexes, has remained unclear. This study focused on determining whether biological sex influences NK cell-mediated immunity against pancreatic cancer *in vitro* and *in vivo*.

In this study, we used the mouse cell line Panc02 as a PDAC model. Although Panc02 was originally described as male,[57] karyotyping and RNA-seq revealed no Y-chromosomal features, indicating the loss of sex-specific properties. Loss of the Y chromosome is a frequent event in human PDAC and in multiple PDAC cell lines.[69] In addition, Panc02 displays SMAD4 deficiency, which is described as a PDAC hallmark,[70] and expresses mild inflammatory features.[57] Therefore, it represents a sex-neutral target suitable for studying NK cell function in PDAC. The Panc02 cells not only proved to be a good model *in vitro* but also *in vivo*. We established a syngeneic allograft mouse model by injecting Panc02 cells subcutaneously into the right leg of male and female BL6 mice. The histological structure of the allograft tumors was remarkably similar to that of human PDAC, as it included a complex tumor stroma infiltrated with immune cells (e.g., TAMs, CAFs, and T and NK cells). A major advantage of the allograft model in comparison with genetically engineered cancer models is the identical origin of the tumor cells, which is consistent across age, sex, and individual properties of the mice. The context of studying sex-dependent differences, therefore, allows maximal focus on infiltrating immune cells and the sex of the host. In addition, in genetically engineered mouse models, tumors do not occur at exactly the same time in all animals, and often progress before the immune system of the host is fully developed. With the allograft model, we could ensure a uniform starting point for all animals, ensuring that each had a fully mature immune system. This approach allowed us to isolate and focus solely on the sex-based differences, free from any time-dependent biases. Future studies should involve an orthotopic injection of Panc02 cells into the pancreas of male and female mice, as this could allow investigations into the impact of tumorigenesis on the exocrine pancreatic function.[71]

In order to study sex-dependent differences in NK cell function, we isolated primary NK cells from male and female BL6 mice to be used in subsequent killing assays. The primary NK cells were phenotypically naïve after isolation, but transcriptomic analysis revealed subtle intrinsic sex-dependent differences in gene regulation. We found that male NK cells expressed lower levels of H3K27me3-regulated genes, a difference likely driven by reduced UTX activity. UTX was shown to influence NK cell cytotoxicity in a sex-dependent manner.[25] As an X-linked gene that escapes inactivation, UTX is expressed at higher levels in females,[25] facilitating the removal of repressive H3K27me3 and supporting sex-specific gene regulation. The H3K27me3 pathway is known to regulate the activity of various immune cells in the context of cancer.[72] In addition, downregulated cytokine response signatures and upregulation of an IL-4/IL-10 program could be observed, suggesting a more immunosuppressive baseline in male NK cells.[73] Our findings are consistent with previous reports that describe an enhanced intrinsic cytotoxic capacity in female NK cells.[14,25]

During cultures with and without Panc02 cells, we showed that IL-2 alone was capable of activating NK cells. In turn, Panc02 cells dampened this activation, which was reflected in diminished surface densities of the activating receptors NCR1 and NKG2D. Tumor cells employ various strategies to escape NK cell-mediated responses; among these, the shedding of ligands to impair NKG2D-mediated activation is a well-characterized mechanism of immune evasion.[74] NK cell maturation was evident in the killing assays, with female NK cells displaying a slightly more advanced differentiation state in co-culture. Furthermore,

female NK cells respond more strongly to Panc02, and the enhanced immune response is driven predominantly by tumor contact.[54,73] In contrast, male NK cells are largely IL-2-driven and less responsive to tumor cell-derived cues. These results were validated with another PDAC cell line (KPC3595), albeit the differences were pronounced to a lesser extent because of its lower suitability as a killing target. Functional killing assays with Panc02 cells revealed that female NK cells are able to kill more tumor cells, resulting in slower tumor cell proliferation and higher tumor cell death over time. Importantly, the enhanced killing capacity of female NK cells was independent of sex hormones. We observed slight initial differences in hormone responses between freshly isolated male and female NK cells, which subsided after cultivation in the absence of sex hormones. Notably, the differences in killing efficacy persisted, suggesting that systemic hormonal signals may influence NK cells under physiological conditions but not after isolation in our setting. Importantly, these findings were consistent across both primary NK cells and NK cell line-based models using NK-92, supporting the conclusion that the observed enhanced cytotoxicity of female NK cells can be attributed to stable epigenetic programs rather than hormonal influences.

The observed sex differences *in vitro* correlate with findings from the *in vivo* allograft setting, reinforcing the relevance of NK cell-tumor interactions. Female tumors grow more slowly, which is correlated with a higher NK cell infiltration. In line with these findings, previous studies have demonstrated that the NK cell infiltration of tumors from PDAC patients is positively correlated with survival.[30,40] A decrease in peripheral NK cells was evident in female allografts, which was most likely caused by enhanced NK cell migration to the tumor site. Following up from here, multiplex spatial immunofluorescence staining and RNA-seq revealed differences in the overall tumor composition between males and females. Male tumors display a dense, compact architecture, including stiff and tumor-promoting stroma, increased basal-like properties, and increased proliferative properties. The male TME is characterized by the infiltration of immune-suppressive cell types, and NK cells are characterized by reduced infiltration, activation, and cytotoxic function. In contrast, female tumors showed a looser architecture, with less basal-like properties and reduced proliferation, leading to an overall immune-permissive stroma with stronger immune response programs and infiltration of pro-inflammatory immune cells. The infiltrating NK cells were strongly activated and displayed high cytotoxic activity. It is reasonable to assume that the loose, pro-inflammatory tumor structure in female mice favors and facilitates NK cell infiltration, as well as other immune cells of the TME. In this context, intrinsic differences in NK cell function collaborate with an environment conducive to NK cells, leading to enhanced NK cell immunity in females *in vivo*. It must be said that, despite NK cells constitute only a small fraction of the TME, they present some of the largest sex differences in terms of their immigration and activation among infiltrating immune cells. In addition, their strong sex-dependent regulation might be far more relevant in the context of naturally arising tumors than in the high-dose allograft injection model, as they are crucial for the control of single transformed cells and metastases.[2] The data from this study support the conclusion that NK cells provide a sex-specific contribution to anti-tumor immunity in concert with the TME and other immune cells. Nevertheless, future NK cell-depletion studies will be needed to definitively delineate their causal role. Beyond this, the greater pro-inflammatory TME of female mice, together with enhanced PD-L1 expression on tumor cells, suggests a greater responsiveness of female allograft tumors to ICB, which was further supported by the GSEA data. This suggests that sex contributes to the prognostic outcome of immune therapies but likely acts in concert with additional influencing factors.[64] Future studies should therefore prioritize the analysis of clinical data from male and female patients undergoing immunotherapies in order to determine whether sex-specific differences are reflected in patient outcomes.

Large-scale pan-cancer bulk transcriptomic studies of human tumors have provided important insights into sex-associated immune differences and offer a useful foundation for translational research.[36] However, bulk analyses can mask sex-specific effects within individual immune cell populations. Future studies combining single-cell and spatial transcriptomic approaches may therefore be needed to more precisely define sex-specific immune programs in the tumor microenvironment. Such analyses could help clarify whether female PDAC patients differentially benefit from immune-based combination therapies when stratified by sex-associated immune features, as suggested by mouse data.

Although we proved that initial sex-dependent differences in NK cell cytotoxicity are independent of sex hormones *in vitro*, they must be considered in the *in vivo* setting. We did not monitor the

estrous cycle in female mice in order to assess general sex differences under standard laboratory conditions. Additionally, including females at various cycle stages reflects natural biological variability. It is worth noting that hormone levels in females can vary more extensively across different life stages than during the normal estrous cycle, and the variability introduced by the estrous cycle stage in female mice is generally not greater than the variability observed among males.[75] Nevertheless, hormones may influence tumor architecture *in vivo*. In allograft tumors, we observed slightly increased AR and ER responses in males. It should be noted that both estrogen and androgen response gene sets include numerous genes involved in cell proliferation. Considering our data showing that male tumors grow faster, this may therefore reflect, at least in part, variations in tumor proliferation rates rather than direct effects of sex hormones. Nevertheless, future studies should involve allograft models using gonadectomized mice to elucidate whether the observed differences in tumor composition persist independently of sex hormones.

Together, our study offers relevant insights into the sex-dependent regulation of NK cells in PDAC. These findings indicate that the combination of initial enhanced NK cell function and NK cell-beneficial tumor architecture in females results in superior NK cell-mediated tumor control in patients with PDAC. This occurs within the context of a proinflammatory TME with enhanced immune cell infiltration, potentially explaining the associations between female sex and reduced PDAC incidence and mortality. Although hormones might play a role in *in vivo*, the observed *in vitro* differences in NK cell cytotoxicity are most likely attributable to intrinsic (epigenetic) mechanisms. The definitive role of NK cells in our *in vivo* setting remains to be elucidated, as further experiments, such as NK cell depletion experiments or the use of NK cell-deficient mice, are needed to clarify this point. Although the mouse data reveal clear patterns, the extent to which these findings apply to the much more heterogeneous human system still needs to be determined. Sex-based disparities warrant increased consideration in clinical practice and underscore the importance of considering biological sex as a variable in future research and targeted therapy.

Disclosure of potential conflicts of interest

The authors report there are no competing interests to declare.

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Author contributions

CRediT: **Elise Arlt:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Nadine Bley:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Alexander Rausch:** Investigation, Software; **Sandra Hauser:** Conceptualization, Investigation, Methodology, Resources, Writing – review & editing; **Jens Pietzsch:** Conceptualization, Investigation, Methodology, Resources, Writing – review & editing; **Danny Misiak:** Data curation, Formal analysis, Software; **Thomas Liehr:** Investigation, Methodology, Writing – review & editing; **Stefanie Kankel:** Investigation; **Theresa Dautz:** Investigation; **Luisa Dreher:** Investigation; **Xiángméng Kǒng:** Investigation; **Heike Kielstein:** Funding acquisition, Project administration, Resources, Supervision; **Ivonne Bazwinsky-Wutschke:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

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Data availability statement

All data generated in this study are available from the corresponding author upon reasonable request. The flow cytometry data (FCS files) that were generated and analyzed in this study could not be uploaded to a public repository, as flowrepository.org has temporarily disabled the creation of new experiments owing to limited space.

The data created by RNA sequencing are deposited under GEO Accession: GSE309514 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE309514).

The graphics were created with Biorender.com and the Global Cancer Observatory website.

Ethics statement

All animal experiments were carried out according to the guidelines of the German Regulations for Animal Welfare and have been approved by the local Ethical Committee for Animal Experiments. The euthanasia of the mice was conducted in accordance with the guidelines and regulations of the Martin-Luther-University Halle-Wittenberg and was approved by the local animal welfare officer under the number I11M32. The experiments with allograft mice were performed according to the animal testing application of the Helmholtz-Zentrum Dresden-Rossendorf (TVV 48/2023 (25-5131/562/52), Landesdirektion Sachsen, Chemnitz, Germany). The criteria according to the 3Rs (replacement, reduction, and refinement) were considered and applied accordingly. All efforts were made to minimize animal suffering and to use the minimum number of animals required to achieve statistically significant results.

List of abbreviations

AI	Artificial Intelligence
Ar/AR	Androgen Receptor
ASR	Age-standardized rate
BL6 mice	C57BL/6 mice
CAF	Cancer associated fibroblast
DAB	3,3'-Diaminobenzidin
DAP12	DNAX-activating protein 12
DAPI	4′,6-Diamidino-2-phenylindol
DHT	Dihydrotestosterone
DMEM	Dulbecco‘s Modified Eagles Medium
E:T	Effector:Target
Esr/ER	Estrogen Receptor
FACS	Fluorescence-activated Cell Sorting
FBS	Fetal bovine serum
FFPE	Formalin-Fixed Paraffin-Embedded
FMO	Fluorescence Minus One
GFP	Green Fluorescent Protein
GM-CSF	Granulocyte Macrophage Colony Stimulating Factor
GOBP	Gene Ontology Biological Process
GPR	G-protein-coupled receptor
GSEA	Gene Set Enrichment Analysis
GZMB	Granzyme B
H.E.	Haematoxylin-Eosin
H3K27	Histone H3 Lysine 27 trimethylation
iCAF	Inflammatory CAF
ICB	Immune Checkpoint Blockade

IFN	Interferon
IGHG1	Immunglobulin gamma 1 Chain C Region
IL	Interleukin
ITAM	Immunoreceptor Tyrosine-based Activation Motif
JAK/STAT	Janus Kinase/Signal Transducer and Activator of Transcription
Kdm5	Lysine Demethylase 5
Kdm6	Lysine Demethylase 6
MACS	Magnetic-associated cell sorting
MDSC	Myeloid-Derived Suppressor Cells
MEF	Mouse embryonic fibroblasts
mFISH	Multiocolor Fluorescence in situ Hybridization
MHC	Major Histocompatibility Complex
MICS	Multiplex imaging and cyclic staining
MOI	Multiplicity of infection
myCAF	Myofibroblastic CAF
NES	Normalized Enrichment Score
NF-κB	Nuclear Factor Kappa-light-chain-enhancer of Activated B-cells
NK cell	Natural killer cell
NKT cell	Natural killer T cell
NPC	Neural progenitor cell
PBS	Phosphate Buffered Saline
PD-1	Programmed cell death protein 1
PD-L1	Programmed Death-Ligand 1
PDAC	Pancreatic Ductal Adenocarcinoma
PI	Propidium Iodide
RIN	RNA Integrity Number
RNA-seq	RNA Sequencing
SMAD4	Mothers Against Decapentaplegic Homolog 4
TAM	Tumor associated macrophage
TCGA	The Cancer Genome Atlas
TME	Tumor microenvironment
TNF	Tumor Necrosis Factor
TRAIL	Tumor Necrosis Factor Related Apoptosis Inducing Ligand
UTX	Ubiquitously Transcribed Tetratricopeptide Repeat, X-linked
UTY	Ubiquitously Transcribed Tetratricopeptide Repeat, Y-linked
ZIP9	Zinc transporter 9

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